Machine Learning A

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Preparatory Exercises 1

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1 Thanks for the Fish

Background

Consider the data in the files `smallFishInput.dt` and `smallFishLabel.dt`. They represent the length of a species of fish as a function of age and water temperature. The fish were kept in tanks at 25, 27, 29 and 31 degrees Celsius. After birth, a test specimen was chosen at random every 14 days and its length was measured. The dependent (output) variable in `smallFishLabel.dt`. is the length of the fish. The inputs are the age of the fish in days and the water temperature in degrees Celsius, stored in the corresponding lines in `smallFishInput.dt`. The data are described in the textbooks by Freund and Minton (1979) and Späth (1991). Of course, the length of a fish is a non-linear function of the animal's age. It saturates when the fish is full-grown. In this exercise, we consider the period of growth and investigate if the growth in length can be modelled by a linear function. Therefore, animals older than 90 days were discarded from the original data set.

Task: Linear regression

The goal of modeling is to find a mapping $f : \mathbb{R}^2 \to \mathbb{R}$. The predictor (input) variables are in `smallFishInput.dt` and the response (output) variable in `smallFishLabel.dt`.

Build an *affine* linear model of the data using linear regression. Report the three parameters of the model as well as the mean-squared-error of the model computed over the complete data set.

Compare the mean-squared-error with the variance of the data. Discuss based on this comparison whether you got a good fit or not.

Deliverables: description of software used; mean-squared error; parameters of the regression model; brief discussion relating mean-squared-error to the biased sample variance

2 K-NN: Warm-up

You are given 5 labeled data points, each with a label 'blue' or 'red', summarized as follows:

(x, y)	$(-1, 5)$	$(2.5, 1)$	$(0, 5)$	$(1, 4)$	$(6, -2)$
Label	blue	blue	red	red	blue

(i) What label does the 1-NN classifier predict for $Y = (-0.3, 3.5)$? Explain why.

(ii) Repeat part (i) with 3-NN and 5-NN classifiers.

(iii) Compare the classifiers above qualitatively.

3 Decision Boundary of K-NN Classifier

You are given 5 data points (see Figure 1): $(-1, 0), (2, 1), (2, -2)$ with label $-$, and $(0, 0), (1, 0)$ with label $+$.

(i) Draw the decision boundaries for 1-NN classifier on the figure shown below. Make it accurate enough so that one can tell whether the integer-valued coordinate points in the diagram are on the boundary.

(ii) Consider a new point $(1, -1.01)$. What class does 1-NN predict for this new point? Explain why.

(iii) What class does 3-NN predict for $(1, -1.01)$? Explain why.

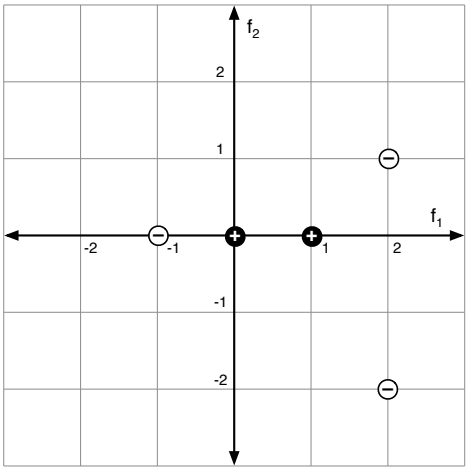


Figure 1: Data points for Question 3

References

R. J. Freund and P. D. Minton. *Regression Methods: A Tool for Data Analysis*, page 111. Marcel Dekker, 1979.

Helmut Späth. *Mathematical Algorithms for Linear Regression*, page 305. Academic Press, 1991.